Blessed G/Michael Catholic Secondary School

2nd Semester Chemistry test two for Grade 10th

May, 2018 E.C Time Allowed: 1:10hr

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28 ½ 30

I. Choose the best answer among the given alternatives.

B 1. A salt that shows efflorescence (loses water of crystallization) is:

A. NaCl B. $CuSO_4.5H_2O$ C. $NaNO_3$ D. NH_4NO_3

C 2. Which of the following salts is derived from a strong acid and a weak base?

A. NaCl B. CH_3COONa C. NH_4Cl D. K_2SO_4

C 3. Which of the following is NOT a property of most solid salts?

A. High melting point C. Good electrical conductor in solid state

B. Soluble in water D. Crystalline structure

C 4. Which gas is released when an acid reacts with a metal carbonate?

A. Hydrogen B. Oxygen C. Carbon dioxide D. Chlorine

C 5. Which of the following salts is generally insoluble in water?

A. $NaNO_3$ B. KCl C. $CaCO_3$ D. $MgNO_3$

C 6. Metallic conduction is primarily due to the movement of:

A. Ions B. Molecules C. Electrons D. Atoms

B 7. What is the effect of increasing temperature on metallic conductivity?

A. Increases C. remains same

B. Decreases D. First increases then decreases

C 8. Which of the following is not a characteristic of electrolytic conductors?

A. They decompose due to the passage of current.

B. Conductivity is due to the movement of ions.

C. Their conductivity decreases with an increase in temperature.

D. Conductance is observed in solutions or molten states.

C 9. Which of the following is not a conductor of electricity?

A. NaCl (aqueous) B. NaCl (molten) C. NaCl (solid) D. Ag (metal)

C 10. Which of the following best describes a substance that produces many ions when dissolved in water resulting in high electrical conductivity?

A. Weak electrolyte B. Nonelectrolyte C. Strong electrolyte D. Covalent compound

B 11. Which of the following is a nonelectrolyte?

A. NaCl B. Sugar(sucrose) C. HCl D. $CuSO_4$

C 12. Why does sodium chloride solution conduct electricity, while a sugar solution does not?

A. Sodium chloride contains more atoms.

B. Sugar is an ionic compound.

C. Sodium chloride dissociates into mobile ions, while sugar does not.

D. Sugar is a stronger electrolyte.

A 13. Which of the following will typically behave as a nonelectrolyte when dissolved in water?

A. Ethanol B. Calcium chloride C. Nitric acid D. Potassium iodide

B 14. Which of the following is considered a "strong base" and thus a strong electrolyte?

A. Ammonia B. Barium hydroxide C. Glucose($C_6H_{12}O_6$) D. Carbonic acid

C 15. Which substance, when added to water, would not produce a conductive solution?

A. Salt (NaCl) B. Vinegar(CH_3COOH) C. Carbon Dioxide D. Lye (NaOH)

B 16. Which of the following solutions cannot conduct electricity?

A. NaCl in water B. Sugar in water C. KCl in water D. $MgCl_2$ in water

A 17. During the electrolysis of molten NaCl, where does the oxidation take place?

A. At the anode B. At the cathode C. In the electrolyte solution D. It does not occur

A 18. Which of the following statements about an electrolytic cell is FALSE?

A. It converts chemical energy into electrical energy.

B. It requires an external power source to drive a non-spontaneous reaction.

C. The anode is connected to the positive terminal of the power source.

D. Reduction occurs at the cathode.

C 19. What is the role of a salt bridge in a galvanic cell?

A. To increase the electrical resistance. C. To maintains charge neutrality in the half-cells

B. To allow electrons to flow between half-cells. D. To heat the solution

A 20. In an electrolytic cell containing molten sodium chloride, what is produced at the cathode?

A. Sodium metal B. Chlorine gas C. Hydrogen gas D. Oxygen gas

B 21. A galvanic cell (like the Daniell cell) differs from an electrolytic cell in that a galvanic cell:

A. Requires an external battery. C. Uses a non-spontaneous redox reaction.

B. Has a negative anode. D. Does not require a salt bridge

A 22. When a lead storage battery is discharged, which of the following processes occurs?

A. Lead sulphate is consumed. C. Lead is formed at the anode.

B. Sulphuric acid is consumed. D. Water is consumed.

C 23. What is the function of a salt bridge in a galvanic cell?

A. To mix the two solutions. C. To maintain electrical neutrality in the half-cells.

B. To increase the cell voltage. D. To produce electrons.

B 24. In a galvanic cell, electrons flow through the external circuit from:

A. Cathode to Anode B. Anode to Cathode C. Salt bridge to Cathode D. Salt bridge to Anode

B 25. In a Daniel cell, what happens to the mass of the Zn electrode?

A. It increases B. It decreases.. C. It remains constant. D. It dissolves and then redeposit

C 26. What is the defining characteristic of a secondary cell?

A. It is non-rechargeable.

B. The electrochemical reactions are irreversible.

C. The redox reaction is reversible, allowing it to be recharged.

D. It can only be used once and then discarded

D 27. Which of the following is an example of a secondary cell?

A. Leclanché cell B. Dry cell C. Mercury cell D. Lead-acid battery

C 28. What is the main component of the electrolyte in a commonly used automotive lead-acid storage battery?

A. Potassium hydroxide B. Ammonium chloride C. Sulphuric acid D. Zinc chloride

C 29. Which of the following best describes an endothermic reaction?

A. Heat is released to the surroundings.

B. The enthalpy of products is lower than reactants.

C. ΔH is positive, and the surroundings cool down.

D. The reaction vessel feels warm to the touch

B 30. What is the composition of the electrolyte in a standard dry Leclanché cell?

A. Dilute Sulphuric Acid B. Ammonium Chloride Paste C. Sodium Hydroxide D. Copper Sulphate

Bonus

Write the anode cathode and the cell reaction during the charging process of a lead acid battery